

Sigmoid function: $f(x) = \frac{1}{1+e^{-x}}$

Y function: $y_n = f(\sum w_{ij} \cdot x_i)$

New weight: $\Delta W_{new} = \mu \cdot y_i \cdot \delta + w_{old}, \quad \mu = 0.1$

$$\delta = \begin{cases} y_p(1 - y_p) \cdot (y_f - y_p), & \text{если выходной нейрон} \\ y_p(1 - y_p) \cdot \sum w_{ij} \cdot \delta, & \text{если промежуточный нейрон} \end{cases}$$

Algorithm:

Forward Pass:

$$H_1 = x_1 \cdot w_{11} + x_2 \cdot w_{21}$$

$$y_1 = f(H_1)$$

$$H_2 = x_1 \cdot w_{12} + x_2 \cdot w_{22}$$

$$y_2 = f(H_2)$$

$$H_3 = h_1 \cdot w_{13} + h_2 \cdot w_{23}$$

$$y_3 = f(H_3)$$

Backpropagation:

$$\delta = y_3(1 - y_3)(0.5 - y_3)$$

$$\delta_2 = y_2(1 - y_2)(w_{23})(\delta)$$

$$\delta_1 = y_1(1 - y_1)(w_{13})(\delta)$$

New weights:

$$\Delta W_{new(ij)} = w_{ij} + 0.1 \cdot y_i \cdot \delta$$

Parameters:

Iteration	y3	w11	w12	w21	w22	w13	w23
0	0.682042	0.099100	0.397100	0.797600	0.592600	0.272400	0.873100
1	0.681237	0.099018	0.396830	0.797389	0.591906	0.269717	0.870486
2	0.680435	0.098937	0.396562	0.797181	0.591216	0.267042	0.867882
3	0.679634	0.098857	0.396295	0.796976	0.590530	0.264375	0.865285
4	0.678835	0.098779	0.396030	0.796773	0.589848	0.261717	0.862698

Iteration	y3	w11	w12	w21	w22	w13	w23
5	0.678038	0.098701	0.395766	0.796573	0.589170	0.259068	0.860119

Diagram of a neural network with two input nodes (x_1, x_2), two hidden nodes (h_1, h_2), and one output node (y_3). Weights are labeled as w_{ij} where i is the source node and j is the target node.

Handwritten calculations:

$$f(x) = \frac{1}{1 + e^{-x}}$$

$$x_1 = 0,35$$

$$x_2 = 0,9$$

$$w_{11} = 0,0991$$

$$w_{12} = 0,3971$$

$$w_{21} = 0,7921$$

$$w_{22} = 0,5926$$

$$w_{13} = 0,2724$$

$$w_{23} = 0,8731$$

$$y_f = 0,5$$

$$E = (0,5 - 0,682) = -0,182$$

$$y_1 = \sum w_{ij} \cdot x_i = (0,35 \cdot 0,0991) + (0,9 \cdot 0,7921) = 0,034685 + 0,71289 = 0,7525$$

$$f(y_1) = 1 / (1 + e^{-0,7525}) = 0,6797$$

$$y_2 = (x_1 \cdot w_{12}) + (x_2 \cdot w_{22}) = 0,6723 \approx 0,662$$

$$y_3 = (h_1 \cdot w_{13}) + (h_2 \cdot w_{23}) = 0,7631 \approx 0,682$$

1st iteration:

$$y_1 = f(H_1) = f(x_1 \cdot w_{11} + x_2 \cdot w_{21}) = f((0,35 \cdot 0,0991) + (0,9 \cdot 0,7976)) = f(0,7525)$$

$$= \frac{1}{1 + e^{-0,7525}} = 0,6797$$

$$y_2 = f(H_2) = f(x_1 \cdot w_{12} + x_2 \cdot w_{22}) = f((0,35 \cdot 0,3971) + (0,9 \cdot 0,5926)) = f(0,6723)$$

$$= \frac{1}{1 + e^{-0,6723}} = 0,662$$

$$y_3 = f(h_1 \cdot w_{13} + h_2 \cdot w_{23}) = f((0,7525 \cdot 0,2724) + (0,6723 \cdot 0,8731)) = f(0,7631)$$

$$= \frac{1}{1 + e^{-0,7631}} = 0,682$$

$$\Delta W_{\text{new}} = \eta \cdot y_i \cdot \delta + W_{\text{old}} \quad \frac{1}{1+e^{-x}} \quad |\eta = 0,1|$$

$$\delta = \begin{cases} \text{если выходной нейрон, } y_p(1-y_p) \cdot (y_f - y_p) \\ \text{если промежуточный нейрон, } y_p(1-y_p) \cdot \sum W_{ij} \cdot \delta_j \end{cases}$$

$$\delta_3 = 0,682(1 - 0,68) \cdot (0,5 - 0,68) = 0,682 \cdot (-0,182) \cdot 0,32 = -0,0406$$

$$\delta_4 = 0,67(1 - 0,67) \cdot 0,27 \cdot (-0,0406) = -0,00265$$

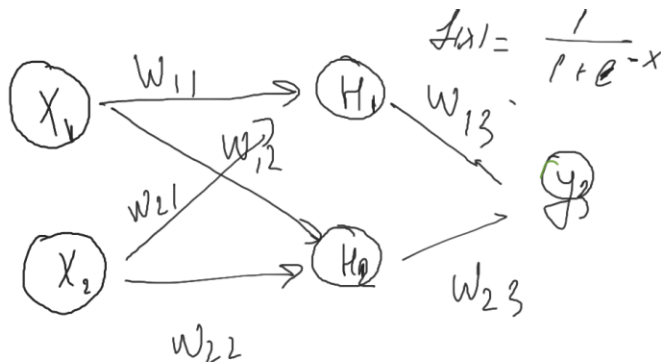
$$\delta_2 = 0,66(1 - 0,66) \cdot 0,87 \cdot (-0,0406) = -0,0082$$

$$\Delta W_{13} = 0,1 \cdot 0,6797 \cdot (-0,0406) + 0,27 = 0,2672$$

$$\delta = 0,682(1 - 0,68) \cdot (0,5 - 0,68) = 0,682(-0,182) \cdot (-0,18) = -0,0406$$

$$\delta_1 = 0,67(1 - 0,67) \cdot 0,27(-0,0406) = -0,00265$$

$$\delta_2 = 0,66(1 - 0,66) \cdot 0,87(-0,0406) = -0,0082$$



$$x_1 = 0,35$$

$$x_2 = 0,9$$

$$W_{11} = 0,0991$$

$$W_{12} = 0,3971$$

$$W_{21} = 0,7926$$

$$W_{22} = 0,5926$$

$$W_{13} = 0,2729$$

$$W_{23} = 0,8731$$

$$y_f = 0,5$$

$$E = (0,5 - 0,682) = -0,182$$

$$y_1 = \sum W_{ij} \cdot x_i = (0,35 \cdot 0,0991) + (0,9 \cdot 0,7926) = 0,034685 + 0,71784 = 0,7525$$

$$f(y_1) = 1 / (1 + e^{-0,7525}) = 0,6797$$

$$y_2 = (x_1 \cdot W_{12}) + (x_2 \cdot W_{21}) = 0,6723 \approx 0,662$$

$$y_3 = (h_1 \cdot W_{13}) + (h_2 \cdot W_{23}) = 0,7631 \approx 0,682$$

$$\Delta W_{13} = 0,1 \cdot 0,6797 \cdot (-0,0406) + 0,27 = 0,2672$$

$$\Delta W_{23} = 0,1 \cdot 0,662 \cdot (-0,0406) + 0,8731 = 0,87041228$$

$$\Delta W_{11} = 0,1 \cdot 0,6797 \cdot (-0,0406) + 0,0991 = 0,096340418$$

$$\Delta W_{21} = 0,1 \cdot 0,662 \cdot (-0,0406) + 0,7976 = 0,79491228$$

$$\Delta W_{12} = 0.1 * 0.6797 * (-0.0406) + 0.3971 = 0.394340418$$

$$\Delta W_{22} = 0.1 * 0.662 * (-0.0406) + 0.5926 = 0.5899122800000001$$

Iteration 1

Forward pass:

$$H_1 = 0.35 \times 0.0991 + 0.9 \times 0.7976 = 0.752525 \rightarrow y_1 = \text{sigmoid}(0.752525) = 0.679729$$

$$H_2 = 0.35 \times 0.3971 + 0.9 \times 0.5926 = 0.672325 \rightarrow y_2 = \text{sigmoid}(0.672325) = 0.662024$$

$$H_3 = 0.679729 \times 0.2724 + 0.662024 \times 0.8731 = 0.763171 \rightarrow y_3 = \text{sigmoid}(0.763171) = 0.682042$$

$$\delta_3 = y_3(1-y_3)(0.5-y_3) = 0.682042 \times 0.317958 \times (-0.182042) = -0.0394777$$

$$\delta_1 = y_1(1-y_1) \times w_{13} \times \delta_3 = 0.679729 \times 0.320271 \times 0.2724 \times (-0.0394777) = -0.0023411$$

$$\delta_2 = y_2(1-y_2) \times w_{23} \times \delta_3 = 0.662024 \times 0.337976 \times 0.8731 \times (-0.0394777) = -0.0077122$$

New weights:

$$w_{13} = 0.2724 + 0.1 \times 0.679729 \times (-0.0394777) = 0.2724 - 0.0026834 = 0.269717$$

$$w_{23} = 0.8731 + 0.1 \times 0.662024 \times (-0.0394777) = 0.8731 - 0.0026135 = 0.870486$$

$$w_{11} = 0.0991 + 0.1 \times 0.35 \times (-0.0023411) = 0.0991 - 0.0000819 = 0.099018$$

$$w_{21} = 0.7976 + 0.1 \times 0.9 \times (-0.0023411) = 0.7976 - 0.0002107 = 0.797389$$

$$w_{12} = 0.3971 + 0.1 \times 0.35 \times (-0.0077122) = 0.3971 - 0.0002699 = 0.396830$$

$$w_{22} = 0.5926 + 0.1 \times 0.9 \times (-0.0077122) = 0.5926 - 0.0006941 = 0.591906$$

Iteration 2

Forward:

$$H_1 = 0.35 \times 0.099018 + 0.9 \times 0.797389 = 0.752307 \rightarrow y_1 = 0.679681$$

$$H_2 = 0.35 \times 0.396830 + 0.9 \times 0.591906 = 0.671606 \rightarrow y_2 = 0.661863$$

$$H_3 = 0.679681 \times 0.269717 + 0.661863 \times 0.870486 = 0.759464 \rightarrow y_3 = 0.681237$$

$$\delta_3 = -0.0393562$$

$$\delta_1 = -0.0023110$$

$$\delta_2 = -0.0076672$$

New weights:

$$w_{13} = 0.269717 - 0.0026750 = 0.267042$$

$$w_{23} = 0.870486 - 0.0026048 = 0.867882$$

$$w_{11} = 0.099018 - 0.0000809 = 0.098937$$

$$w_{21} = 0.797389 - 0.0002080 = 0.797181$$

$$w_{12} = 0.396830 - 0.0002684 = 0.396562$$

$$w_{22} = 0.591906 - 0.0006900 = 0.591216$$

Iteration 3

Forward:

$$H_1 = 0.752091 \rightarrow y_1 = 0.679634$$

$$H_2 = 0.670891 \rightarrow y_2 = 0.661703$$

$$H_3 = 0.755770 \rightarrow y_3 = 0.680435$$

$$\delta_3 = -0.0392343$$

$$\delta_1 = -0.0022812$$

$$\delta_2 = -0.0076223$$

New weights:

$$w_{13} = 0.264375$$

$$w_{23} = 0.865285$$

$$w_{11} = 0.098857$$

$$w_{21} = 0.796976$$

$$w_{12} = 0.396295$$

$$w_{22} = 0.590530$$

Iteration 4

Forward:

$$H_1 = 0.751878 \rightarrow y_1 = 0.679588$$

$$H_2 = 0.670180 \rightarrow y_2 = 0.661543$$

$$H_3 = 0.752090 \rightarrow y_3 = 0.679634$$

$$\delta_3 = -0.0391120$$

$$\delta_1 = -0.0022516$$

$$\delta_2 = -0.0075776$$

New weights:

$$w_{13} = 0.261717$$

$$w_{23} = 0.862698$$

$$w_{11} = 0.098779$$

$$w_{21} = 0.796773$$

$$w_{12} = 0.396030$$

$$w_{22} = 0.589848$$

Iteration 5

Forward:

$$H_1 = 0.751669 \rightarrow y_1 = 0.679542$$

$$H_2 = 0.669473 \rightarrow y_2 = 0.661385$$

$$H_3 = 0.748424 \rightarrow y_3 = 0.678835$$

$$\delta_3 = -0.0389893$$

$$\delta_1 = -0.0022221$$

$$\delta_2 = -0.0075329$$

New weights (final):

$$w_{13} = 0.259068$$

$$w_{23} = 0.860119$$

$$w_{11} = 0.098701$$

$$w_{21} = 0.796573$$

$$w_{12} = 0.395766$$

$$w_{22} = 0.589170$$